

Definition of the Derivative

Date_____ Period____

Use the definition of the derivative to find the derivative of each function with respect to x .

1) $y = 2x + 2$

2) $y = x + 5$

3) $y = 2x + 5$

4) $y = 5x + 4$

5) $y = 4x^2 + 4$

6) $y = -3x^2 + 2$

7) $y = -4x^2 + 1$

8) $y = x^2 - 5$

9) $f(x) = x^2 + x - 1$

10) $f(x) = 4x^2 + 4x - 3$

Critical thinking question:

- 11) Look at your answers for problems 1-10. Try to determine a pattern to guess the derivative of $y = 2x^2 + 3x + 7$.